Syllabus Year: 2024-2025

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - III

Course Code: BCA202-3 **Max Marks: 100**

Course Name: OPERATING SYSTEM **Duration: 03:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Demonstrate an understanding of the basic working process of an operating system.

CO2: Explain the importance of process and scheduling.

CO3: Analyse the issues in synchronization and memory management.

Section A **2*10=20**

1)	Demonstrate how a process transitions between the ready, running, and waiting states in an operating system by providing a brief explanation and identifying the triggering events for each state change.	**2**	(CO1, L3)
2)	Evaluate the effectiveness of the Copy-on-Write and Write-Trough page replacement strategies in terms of memory usage and performance for a multi-process system.	**2**	(CO3, L5)
3)	Identify and state two main categories of operating systems mentioned in your course syllabus.	**2**	(CO1, L1)
4)	Discuss the importance of process scheduling in an operating system.	**2**	(CO2, L2)
5)	Demonstrate how space allocation and deallocation are performed using linked allocation in a file system.	**2**	(CO1, L3)
6)	Identify and state one method used in process scheduling to ensure that all processes get a fair share of CPU time.	**2**	(CO2, L1)
7)	Identify and state two main functions of an operating system related to memory management.	**2**	(CO1, L1)
8)	Analyze the synchronization issues that arise in managing shared memory among concurrent processes and compare these issues with those encountered in file system concurrency management. Discuss how mutual exclusion can be achieved in both contexts.	**2**	(CO3, L4)
9)	State two page replacement algorithms used in operating systems and briefly mention their primary objective.	**2**	(CO3, L1)
10)	Describe the boot process of a computer system, highlighting the roles of BIOS/UEFI and the operating system in this sequence.	**2**	(CO1, L2)

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Section B — 5*6=30

11)	Identify and explain the steps involved in handling a deadlock using the Banker's algorithm with an example.	**5**	(CO1, L1)
12) a)	Identify and explain three key characteristics of a process in an operating system, providing examples for each.	**5**	(CO2, L1)
	OR		
b)	Identify and explain two advantages of using demand paging in process management.	**5**	(CO2, L1)
13)	Discuss how priority scheduling affects system performance with an example.	**5**	(CO2, L2)
14)	Explain the process of context switching and discuss its importance in effective system scheduling with an example.	**5**	(CO2, L2)
15) a)	Construct a scenario where a real-time operating system (RTOS) uses various scheduling policies to handle different types of processes. Demonstrate how these policies affect process execution and system responsiveness, providing specific examples.	**5**	(CO2, L3)
	OR		
b)	Determine and describe two scheduling algorithms that prioritize processes based on their waiting time. Provide an example scenario where each algorithm would be most effective and explain why.	**5**	(CO2, L3)
16)	Evaluate the effectiveness of Round Robin scheduling algorithm in a multitasking environment by discussing its key advantages and disadvantages, and provide an example scenario where it would perform exceptionally well.	**5**	(CO2, L5)

Section C — 10*5=50

17) a)	Identify and state two main functions of a process in an operating system and explain how these functions are essential for effective process management. Additionally, describe one method each of deadlock prevention and deadlock avoidance to handle potential deadlocks.	**10**	(CO2, L1)
	OR		
b)	Identify and state two key differences between multiprogramming and multiprocessing systems. Additionally, explain how scheduling algorithms are crucial in both types of systems to manage processes effectively.	**10**	(CO2, L1)
18)	Demonstrate the implementation of a simple file system structure by constructing an example scenario where a user creates and deletes files in a directory. In your answer, show how the file control block (FCB) is updated during these operations and explain the implications on free-space management. Additionally, discuss any potential issues that may arise due to fragmentation and propose methods for defragmentation.	**10**	(CO3, L3)

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19) a)	Examine the given scenario and analyse the potential issues related to synchronization and memory management that may arise in a multi-threaded environment. Discuss how these issues can affect system performance. (i) Consider a multi-threaded application where multiple threads access shared resources, such as a single queue for task scheduling. Describe two specific scenarios where thread interference could lead to data corruption or incorrect results. (ii) Suggest synchronization mechanisms that could be used to prevent these issues and explain how they work. (iii) In the context of memory management, discuss the challenges of managing virtual memory in a multi-threaded environment, particularly focusing on demand paging. Provide an example of a scenario where demand paging might fail to meet the requirements of a thread.	**10**	(CO3, L4)
	OR		
b)	Analyze and differentiate between three key aspects of page replacement algorithms in the context of demand paging: the impact on system performance, the trade-offs involved, and the scenarios where each algorithm might be most effective. Provide specific examples for each aspect to support your analysis.	**10**	(CO3, L4)
20)	Demonstrate how process creation, scheduling, and termination are handled in an operating system by constructing a scenario where a new application is installed on a computer running a Linux-based OS. In your response, address the following points (i) Explain the steps involved in creating a new process for this application during installation. (ii) Discuss three different scheduling policies that could be used to manage the newly created process and other existing processes, providing an example of when each policy might be most appropriate. (iii) Describe the process termination phase and detail how resources are cleaned up after the application is uninstalled.	**10**	(CO2, L3)
21) a)	Discuss the principles of mutual exclusion in concurrent process synchronization, and explain how two popular methods (i.e., Semaphores and Monitors) achieve this principle without leading to deadlocks.	**10**	(CO3, L2)
	OR		
b)	Discuss the key issues related to synchronization and memory management in operating systems, specifically focusing on process scheduling and page replacement algorithms. Your answer should be structured into three parts: (i) Explain how priority scheduling can lead to starvation and identify methods to prevent it; (ii) Describe two common page replacement policies and discuss their pros and cons; (iii) Illustrate with an example scenario where a deadlock occurs in a system, and outline the steps to avoid such deadlocks.	**10**	(CO3, L2)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create

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